HOUSE PRICE PREDICTION USING LINEAR REGRESSION

Introduction : Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions. House price prediction is a common application of Machine Learning that helps estimate property values based on different factors. Here, Linear Regression is used to predict house values using the California Housing dataset. The project involves data analysis, visualization, model training and evaluation using Python.

Project Overview: The objective of this project is to build and evaluate a Linear Regression model that predicts median house values using information about California housing districts. The project follows a basic machine-learning workflow: load the dataset, explore and visualize the data, prepare training and testing sets, train the model, generate predictions, and evaluate its performance.

Dataset Description: The project uses the California Housing dataset available through scikit-learn. It contains 20,640 records and eight input features.

Features	Description
○ MedInc	Median income in the district
○ HouseAge	Median House Age
○ AveRooms	Average number of rooms
○ AveBedrms	Average number of bedrooms
○ Population	District population
○ AveOccup	Average household occupancy
○ Latitude / Longitude	Geographical Location

Tools and technologies Used:

Python: Main programming language
Jupyter Notebook: Writing and executing code.
NumPy: Numerical calculations.
Pandas: Data handling and analysis.
Matplotlib: Data visualization.
Scikit-learn: Model building, training and evaluation.
Joblib: Saving and loading the trained model.
PyCharm: Development environment

Data Exploration and Visualization: The dataset is explored using Pandas functions to understand its structure, statistical information and missing values. Matplotlib is used to visualize house value distribution through a histogram and examine the relationship between median income and house values using a scatter plot.

Data Preprocessing and Model Training : The target column MedHouseVal was separated from the input features. The input matrix X contained the eight predictor columns, while y contained the target values. The data was divided using train_test_split with test_size=0.2 and random_state=42. This produced 16,512 training records and 4,128 testing records, corresponding to an 80:20 split. A Linear Regression model from scikit-learn was created and trained using model.fit(X_train, y_train). During training, the algorithm learned an intercept and one coefficient for each input feature. The trained model then generated predictions for the unseen test features using model.predict(X_test).

Model Evaluation : The model was evaluated by comparing its predictions with the actual target values in the test set. Three standard regression metrics were calculated

Metric	Obtained result	Interpretation
MAE	0.5332	Average absolute error; about $53,320 in target units
RMSE	0.7456	Penalizes larger errors more; about $74,558 in target unit
R^2 Score	0.5758	The model explains about 57.58% of the variation in test-set house values

Actual vs Predicted Visualization:- A scatter plot was created with actual values on the horizontal axis and predicted values on the vertical axis. A dashed diagonal reference line was added to represent perfect predictions. Points closer to this line indicate more accurate predictions; points farther away indicate larger errors. The plot provides a visual way to inspect where the model overpredicts or underpredicts

Results and Discussion: The model achieved an MAE of 0.5332, RMSE of 0.7456 and R^2 score of 0.5758. The results indicate that the model explains approximately 57.58% of the variation in house values. The predictions show that Linear Regression can provide useful estimates, although some differences between actual and predicted values remain.

Model Saving: After successful training and evaluation, the Linear Regression model is saved as a **.pkl** file using the Joblib library. This allows the trained model to be loaded and reused for future predictions without repeating the training process.

Advantages:

Easy to understand and implement.
Requires less computational power.
Useful for predicting continuous values.
Provides a basic understanding of Machine Learning.

Conclusion

This project demonstrates the use of Linear Regression for house price prediction using the California Housing dataset. It provides practical experience in data analysis, visualization, model training, prediction and evaluation using Python.

References

Scikit-learn Official Documentation – https://scikit-learn.org/
Pandas Documentation – https://pandas.pydata.org/
NumPy Documentation – https://numpy.org/
Matplotlib Documentation – https://matplotlib.org/
California Housing Dataset – Scikit-learn Dataset Library.